

Hypertensive urgency and emergency guideline (AMU/SDEC)

Summary

This guideline provides information to help with the assessment, management, safe discharge and follow-up of adults referred to the acute medical unit (AMU) or same-day emergency clinic (SDEC) with hypertensive urgency and hypertensive emergency. This includes **definitions** of hypertensive urgency and emergency, guidance about who needs to be **admitted**, protocols for the **parenteral treatment** of hypertensive emergencies, advice about investigation of **secondary hypertension**, and arrangements for **follow-up** in AMU ambulatory clinic.

Definitions

Hypertensive urgency: severely acutely elevated blood pressure with SBP >180 or DBP >120 without end-organ damage.

Hypertensive emergency: elevated blood pressure with acute end-organ damage (retinal, neurological, cardiac or renal) or acute hypertension-mediated emergency (stroke, aortic dissection, pre-eclampsia/eclampsia/HELLP, phaeochromocytoma, scleroderma crisis).

Acute symptoms that may indicate **hypertension-mediated end-organ damage** include: headache, vertigo, syncope, visual impairment, cognitive impairment, reduced consciousness, shortness of breath, chest pain.

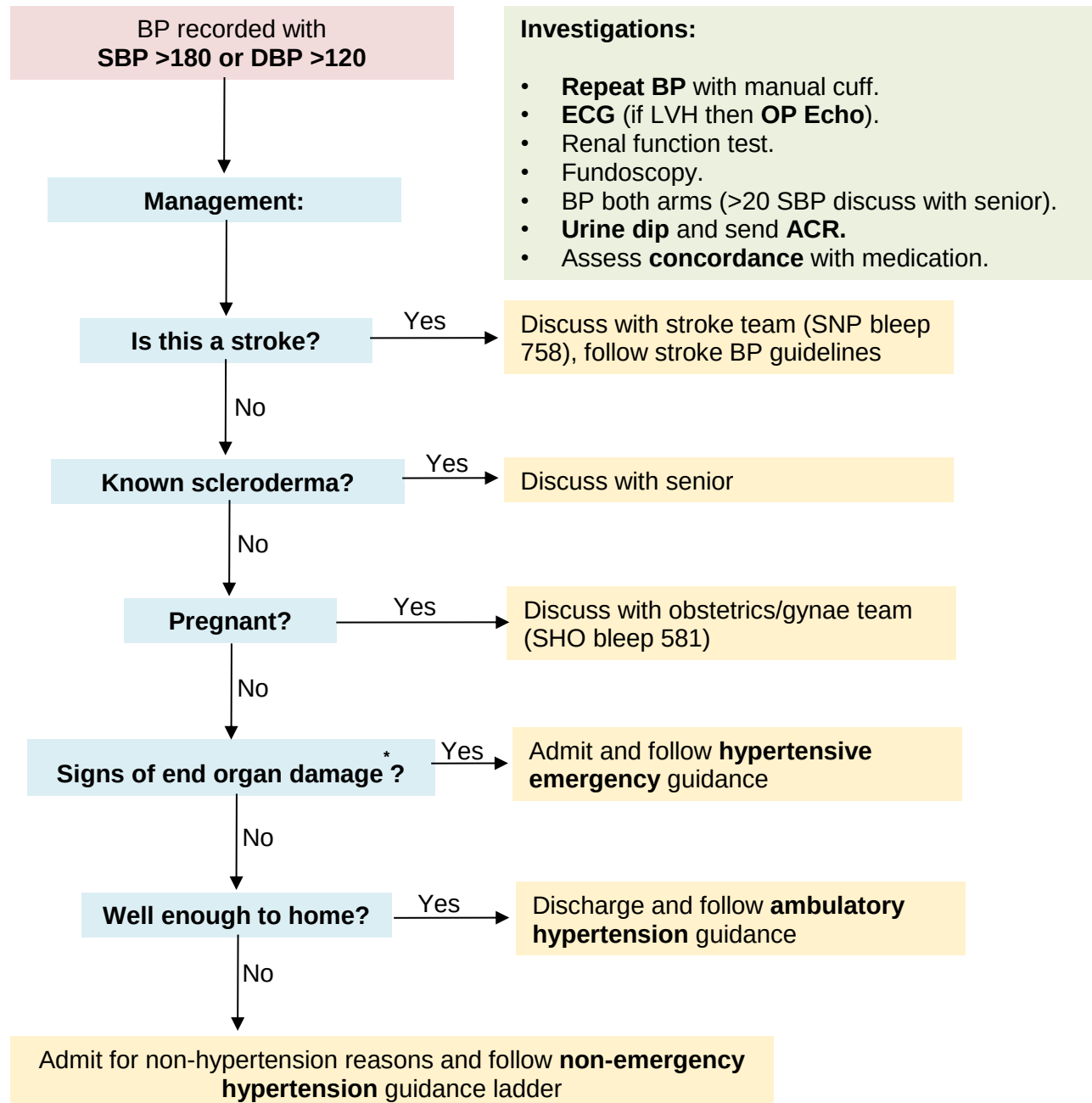
Hypertensive emergencies:

- Hypertensive encephalopathy.
- Aortic dissection.
- Hypertensive myocardial infarction.
- Hypertensive heart failure/pulmonary oedema.
- Hypertensive renal crisis.

Not covered by this guideline:

- Stroke (follow stroke service guideline for BP management in acute stroke).
- Pre-eclampsia, eclampsia and HELLP (discuss urgently with obstetrics team).
- Phaeochromocytoma (discuss urgently with endocrine team/medical seniors).
- Scleroderma crisis (discuss urgently with senior/renal).

Hypertensive Urgency (SBP >180 or DBP >120) pathway:



***Signs of end organ damage:**

- Chest pain with ischaemic ECG or evidence of ACS.
- Pulmonary oedema.
- Acute kidney injury (on blood tests).
- New confusion/encephalopathy.
- Grade 3/4 retinopathy†

† warrants admission for monitoring, but not parenteral treatment.

Non-emergency hypertension guidance ladder

- 1, Low dose calcium channel blocker
(amlodipine 5mg or lercanidipine 10mg)
- 2, Add low dose ACEi/ARB
(ramipril 2.5mg or losartan 50mg)
- 3, Up-titrate CCB and/or ACEi/ARB
 (to maximal tolerated doses)
- 4, Add **indapamide IR 2.5mg**
- 5, If K⁺ <5.0, add **spironolactone 25mg***
 If K⁺ ≥5.0, add **doxazosin IR 1mg** or **carvedilol 12.5mg**

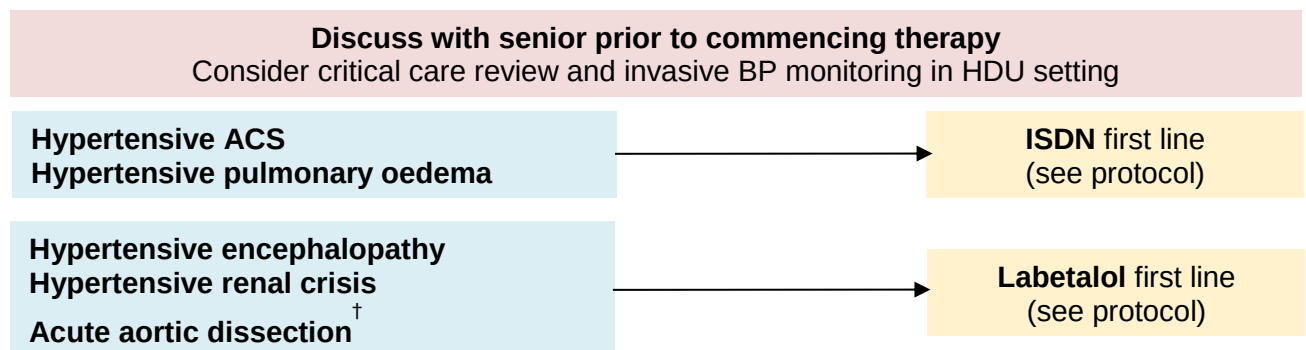
*once on spironolactone, can continue as long as K⁺ remains <5.5.

Notes:

Women of child-bearing age require counselling regarding contraception/future pregnancy as ACEi/ARBs are contra-indicated during pregnancy. Refer to hypertension in pregnancy guidelines if planning to get pregnant.

If ≥80s or frail (e.g. CFS ≥6), titrate slowly aiming BP <160/90. Consider appropriateness of therapy and possibility of postural BP drop.

Hypertensive emergency management



[†]Urgent discussion with vascular surgical team +/- cardiothoracic team in Derriford.

If requiring parenteral BP treatment, cardiovascular hypertensive emergencies should be managed on **CCU**, other hypertensive emergencies should be managed on **Creedy** (stroke patients should go to **Clyst** ward).

New grade 3 or 4 retinopathy without other end-organ damage warrants admission for oral BP lowering, monitoring for ~24 hours and referral for **urgent ophthalmology** review (call 4955).

BP targets

Aortic dissection: SBP <120 and HR <60 as soon as possible.

Hypertensive ACS: SBP <140 as soon as possible.

Hypertensive encephalopathy: SBP reduction by 20-25% immediately and gradually thereafter.

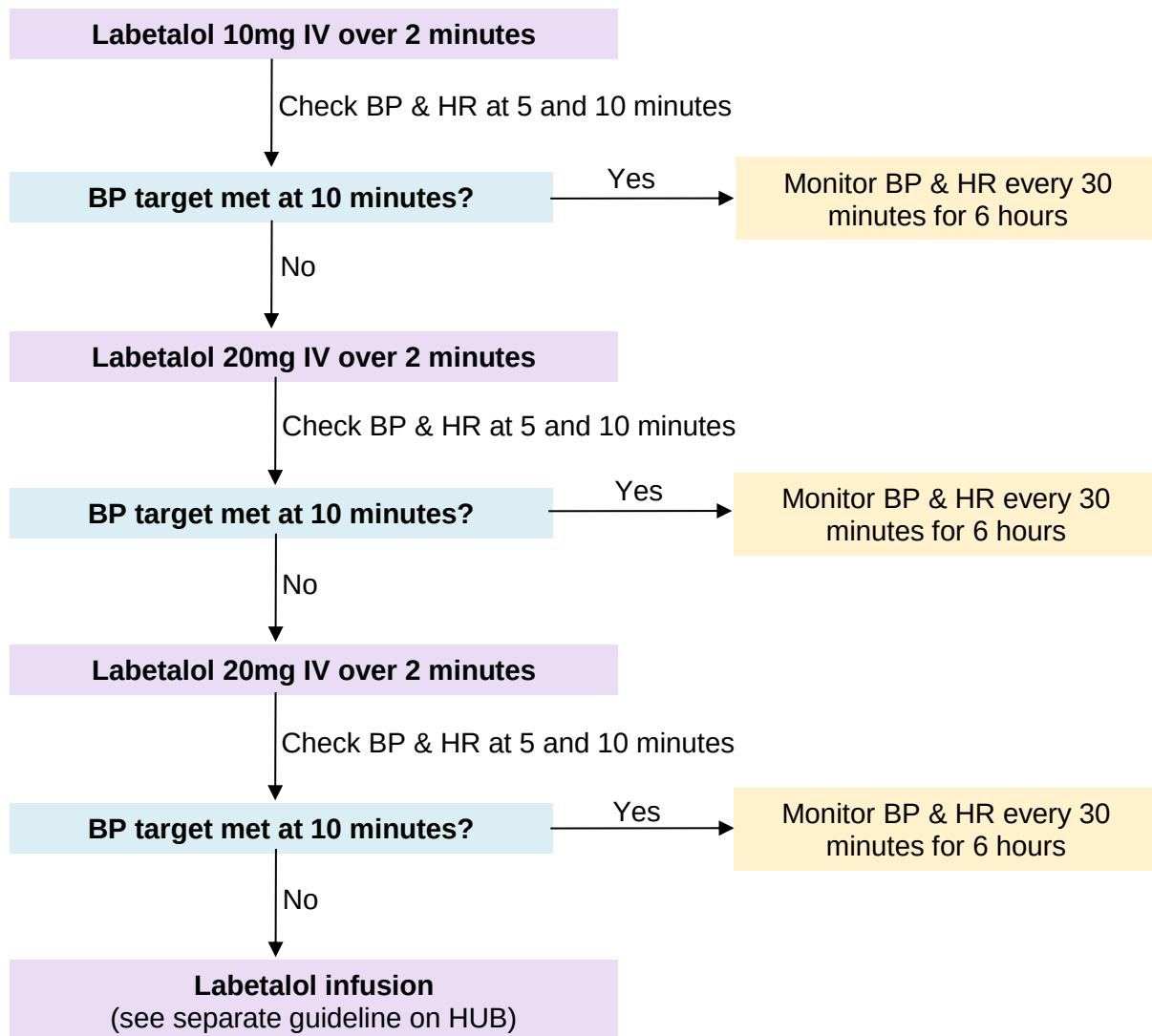
Hypertensive acute pulmonary oedema: SBP <140 as soon as possible.

Other hypertensive emergencies: reduce MAP by 20-25% over several hours.

Labetalol protocol (for use in appropriately skilled areas):

Contra-indications: bronchospasm, hypotension, bradycardia and conduction defects.

Patients should receive **continuous ECG cardiac monitoring**, arrange medical review if bradycardia.



See **Labetalol IV on CCU (Adults)** on HUB and Medusa for further information

Isosorbide dinitrate protocol (for use in appropriately skilled areas):

- Aim for BP targets at timepoints shown on page 3.
- As per trust isosorbide dinitrate protocol – isosorbide dinitrate continuous infusion (1mg/mL).
- **Start infusion at 2mg/hour (2ml/hr).**
- Check BP trend and consider dose adjustments at max 15 minutes intervals to reach target BP.
- **Max rate 10mg/hour (10ml/hr).**
- If BP goal not reached at maximal rate after 30 minutes, discuss with senior medical team.

Dose adjustments:

BP above target → increase by 1-2mg/hr.

BP within 20mmHg below target → continue current infusion rate.

BP is 20-40mmHg below target → reduce infusion rate by 1-2mg/hr.

BP is ≥40mmHg below target → stop infusion.

Alternative nitrates which can be used in place of isosorbide dinitrate per protocol include **sodium nitroprusside** (rarely used on AMU, to be utilised with senior medical oversight, senior nurse to ensure UV protection that comes in the pack is in place) and **glyceryl trinitrate**.

Ambulatory/AHAH pathway:

Referral:

In hours → refer via AHAH team (6107 or find in AMU).

Out of hours → if deemed suitable* for ambulatory pathway by senior.

Management:

- Start oral medications (see non-emergency BP management ladder).
- Advise to buy BP cuff and start completing BP diary (see appendix 3) 72 hours after discharge.
- Advise to visit optician to have retinal pictures for monitoring.
- Refer for ambulatory appointment (within 1 week if >75/CKD/HF, within 2 weeks for others).
- Give AHAH number and card with QR link to NHS website on hypertension.

***Suitability for ambulatory:**

- No end organ damage (including grade 3/4 retinopathy).
- No AKI.
- No chest pain/pulmonary oedema.
- Able to start medications.
- Not pregnant.
- Able to return for ambulatory appointments.

Secondary hypertension screen - consider/screen for when:

- Patients <40 years old with SBP >160 or DBP >100.
- Acute worsening of hypertension in those with previously well-controlled hypertension.
- Resistant hypertension (BP remains >140/90 on 3 agents which include a diuretic).

- SBP >180 or DBP >110.
- Presence of extensive hypertension-mediated end-organ damage.
- Clinical/biochemical features of Cushing's/Conn's/Acromegaly/Thyrotoxicosis.
- Symptoms of pheochromocytoma (episodic headache, sweating, panic attacks, tachycardia, palpitations or family history).
- Obstructive sleep apnoea.

Screen includes:

- Urinalysis and renal function (if not checked already).
- Plasma aldosterone:renin ratio, plasma metanephrines, TFTs.
- Renal USS.
- Echocardiography.

If Cushing's/Acromegaly suspected - discuss with endocrine team.

On review in ambulatory clinic:

Review for hypertensive urgency

↓

SBP <180 and DBP <120 → No → Repeat after 5 minutes with manual cuff (if still high, Mx as HTN urgency)

↓ Yes

If SBP >140/90 (or >160/90 and over 85 or CFS ≥6), increase BP meds by 1 step on ladder
 Review renal function tests if recent ACEi/ARB started*
 Review secondary hypertension screen if sent
 Discharge back to GP care (ask for repeat urine ACR in 3/12)

***ACEi/ARB:** increased incidence of renal artery stenosis in this cohort (other risk factors are new HTN, <40 years, female). Repeat renal function within 2 weeks of started ACEi/ARB is required.

If creatinine rises by >20% then stop ACEi/ARB and up-titrate via other options on ladder.
 Send referral to renal for ?renal artery stenosis and arrange renal USS.

Refer to renal if:

- Microscopic haematuria (in absence of UTI) and urine ACR >30.
- Urine ACR >70.
- ≥20% deterioration in creatinine after ACEi/ARB initiation.

Refer to renal clinic:

On ≥3 BP meds and remains hypertensive.
 Suspected secondary hypertension.

APPENDIX 1: RATIFICATION

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APPENDIX TWO: ACRONYMS/ABBREVIATIONS

ACEi: angiotensin-converting enzyme inhibitor
ACR: albumin creatinine ratio
ACS: acute coronary syndrome
AHAH: acute hospital at home
ARB: angiotensin receptor blocker
BP: blood pressure
CCB: calcium-channel blocker
CFS: clinical frailty score
CKD: chronic kidney disease
DBP: diastolic blood pressure
Echo: echocardiography
GTN: glyceryl trinitrate
HELLP: hemolysis, elevated liver enzymes, and low platelets (in pregnancy)
HF: heart failure
HTN: hypertension
IR: immediate-release
LVH: left-ventricular hypertrophy
OP: outpatient
SBP: systolic blood pressure
SDEC: same day emergency care
SNP: stroke nurse practitioner
TFTs: thyroid function tests
USS: ultrasound scan
UTI: urinary tract infection

APPENDIX 3: PATIENT BLOOD PRESSURE DIARY

Please start monitoring your blood pressure with an automatic blood pressure machine 3 days after your first attendance in hospital with high blood pressure.

We recommend purchasing a blood pressure monitor - there is a list available on the British Heart Foundation website: <https://giftshop.bhf.org.uk/medical-devices/blood-pressure-monitors> which also has guidance about how to choose.

Aim to check your blood pressure 2 times per day, in the morning and in the evening.

Before checking your blood pressure: ensure you are relaxed, with an empty bladder, and have been sitting still for 5 minutes. Aim to avoid exercising, drinking caffeine or smoking for 30 minutes before your reading.

While taking your blood pressure: ensure you are sitting quietly and calmly, with a well size blood pressure cuff on your upper arm, with the cuff at the level of your heart and your arm relaxed.

Take 3 measurements, roughly 1-2 minutes apart each time, and write them in the diary. Please bring the diary with you to your appointments.

Week 1

	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Date							
AM reading 1							
AM reading 2							
AM reading 3							
PM reading 1							
PM reading 2							
PM reading 3							

Week 2

	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Date							
AM reading 1							
AM reading 2							
AM reading 3							
PM reading 1							
PM reading 2							
PM reading 3							

APPENDIX 4: PERIOPERATIVE BP MANAGEMENT CONSIDERATIONS

Peri-operative management has some specific recommendations:

- Newly diagnosed hypertensive patients for elective surgery should be screened for hypertension-mediated end-organ damage and CVD (not necessarily by us).
- Avoid large BP fluctuations peri-op.
- You do not have to defer non-cardiac surgery with grade 1 or 2 hypertension (<180 and/or <110).
- Keep beta-blockers going if on them and do not stop centrally acting drugs such as clonidine.
- Consider transient holding of RAS drugs in patients undergoing non-cardiac surgery).